

Endsem Rubric

①:

$$x_4 - K \leq 0$$

$$L = x_1^2 + x_2^2 + x_3^2 + x_4^2$$

$$+ \lambda (x_1 + x_2 + x_3 + x_4 - 1)$$

$$+ \mu (x_4 - K)$$

$$u \geq 0$$

$$2x_1 + \lambda = 0$$

$$2x_2 + \lambda = 0$$

$$2x_3 + \lambda = 0$$

$$2x_4 + \lambda + u = 0$$

$$x_1 + x_2 + x_3 + x_4 = 1$$

$$x_4 \leq K$$

$$\mu \geq 0$$

$$\mu (x_4 - K) = 0$$

$$x_1 = x_2 = x_3 = -\frac{\lambda}{2}$$

$$\hookrightarrow x_1 = x_2 = x_3 = x_4$$

$$\{ \text{if } x_4 \leq k \text{ and } \mu = 0 \}$$

$$x^* = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4} \right)$$

$$\text{optimal val : } \frac{1}{4}$$

$$\text{if } x_4 = k$$

$$x_1 + x_2 + x_3 = 1 - k$$

$$x_1 = x_2 = x_3$$

$$3x_1 = 1 - k$$

$$x_1 = x_2 = x_3 = \frac{1-k}{3}$$

$$0 \leq k \leq \frac{1}{4}$$

$$x^* = \left(\frac{1-k}{3}, \frac{1-k}{3}, \frac{1-k}{3}, k \right)$$

after putting value :

$$\text{optimal value : } \frac{1 - 2k + 4k^2}{3}$$

(2) :

a) true

as all θ is allowed.

b) false

valuations can be diff.

c) i) true

↳ weierstrass theorem

ii) false

fixed point theorem

d) false

not sufficient for non-convex problems.

(3) :

$$A = \bigcap_{i=1}^{\infty} H_i$$

$$x, y \in H_i$$

$$\theta \in [0, 1]$$

$$\theta x + (1 - \theta) y \in H_i$$

$x, y \in H_i \quad \forall i \in 1 \dots K$
 since every H_i is convex
 $\rightarrow \theta x + (1 - \theta) y \in \bigcap_{i=1}^K H_i = A$
 hence convex

(5) : $\min f_0(x) = 2$
 s.t
 $f_i(x) \leq u_i$
 $i = 1, \dots, m$
 $h_i(x) = v_i$
 $i = 1, \dots, p$

$$L(x, \lambda, v)$$

$$\begin{aligned}
 &= f_0(x) + \sum_{i=1}^m \lambda_i (f_i(x) - u_i) \\
 &\quad + \sum_{i=1}^p v_i (h_i(x) - v_i)
 \end{aligned}$$

$$\lambda_i \geq 0$$

λ_i^*, v_i^*
 $\hookrightarrow$ optimals

$$\frac{d(f_0(\vec{u}))}{du_i} = -\lambda_i^*$$

$$\frac{d(f_0(\vec{u}))}{dv_i} = -v_i^*$$

if u_i increases by $\alpha > 0$

$$u_i + \alpha$$

$$\begin{aligned}\Delta q^* &= \sum_i^m (-\lambda_i^*) \alpha \\ &= -\alpha \sum_i^m \lambda_i^*\end{aligned}$$

decreases

if v_i decreases by $\beta > 0$

$$v_i - \beta$$

$$\begin{aligned}\Delta q^* &= \sum_i^p (-v_i^*) (-\beta) \\ &= \beta \sum_i^p v_i^*\end{aligned}$$

increases

$$\text{new } q^* = \sum_i^p v_i^* - \alpha \sum_i^m \lambda_i^*$$

$$q' = q + \frac{1}{1}$$

④:

$$\min V(f(x))$$

$$s.t. \quad x_1 \geq 0$$

$$x_1 \geq 0$$

$$x_2 \leq 2$$

$$x_2 \leq 2$$

$$b) \quad f(x) = \{(x_1, x_2) \mid x_1^2 + x_2^2 = K, K \geq 0\}$$

$$x_1 + x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

minimal for circle will
be point (0,0) → point



points optimal as $x_1 + x_2 \leq 2$,
if we ↑ any 1, other
gets adjusted acco.